

EECS 498-010 Principles of Machine Learning

Probability Review¹

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¹Reference: CS229 Review of Probability Theory

Probability Review

- Elements of Probability
- Random Variables
- Two Random Variables
- Multiple Random Variables
- Open questions if time left

Elements of Probability

- Sample space Ω : The set of all the outcomes of a random experiment. Here, each outcome $\omega \in \Omega$ can be thought of as a complete description of the state of the real world at the end of the experiment.
- Set of events (or event space) $\mathcal{F}$: A set whose elements $A \in \mathcal{F}$ (called events) are subsets of Ω (i.e., $A \subseteq \Omega$ is a collection of possible outcomes of an experiment).
- Probability measure: A function $P : \mathcal{F} \rightarrow \mathbb{R}$ that satisfies the following properties,
 - $P(A) \geq 0$, for all $A \in \mathcal{F}$
 - $P(\Omega) = 1$
 - If $A_1, A_2, \dots$ are disjoint events (i.e., $A_i \cap A_j = \emptyset$ whenever $i \neq j$), then

$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i)$$

Elements of Probability

Properties:

- If $A \subseteq B \implies P(A) \leq P(B)$.
- $P(A \cap B) \leq \min(P(A), P(B))$.
- $P(\Omega \setminus A) = 1 - P(A)$.
- (Union Bound) $P(A \cup B) \leq P(A) + P(B)$.
- (Inclusion-Exclusion Principle) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- (Law of Total Probability) If $A_1, \dots, A_k$ are a set of disjoint events such that $\cup_{i=1}^k A_i = \Omega$, then $\sum_{i=1}^k P(A_i) = 1$.

Elements of Probability: Total Probability Theorem

- Event Space: A set of disjoint events that form a partition of the sample space (Ω) .
- For any Event Space $\{B_1, B_2, \dots, B_m\}$,

$$P(A) = \sum_{i=1}^m P(A \cap B_i).$$

Elements of Probability: Conditional Probability

Let B be an event with non-zero probability. The conditional probability of any event A given B is defined as,

$$P(A \mid B) \triangleq \frac{P(A \cap B)}{P(B)}$$

- $P(A \mid B)$ is the probability measure of the event A after observing the occurrence of event B .
- Two events are called independent if and only if $P(A \cap B) = P(A)P(B)$ (or equivalently, $P(A \mid B) = P(A)$).
- Independence is equivalent to saying that observing B does not have any effect on the probability of A .

Elements of Probability: Independence

Example: A family has two children. What is the probability that both are boys given at least one is a boy?

Elements of Probability: Exclusivity and Independence

Example: Consider the random experiment of rolling a die. Let A denote the event that an odd number is rolled while B represents the event that an even number is rolled. Are A and B independent events?

Probability Review

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Random Variables

A random variable X is a function $X : \Omega \longrightarrow \mathbb{R}$.² Typically, we will denote random variables using upper case letters $X(\omega)$ or more simply X (where the dependence on the random outcome ω is implied). We will denote the value that a random variable may take on using lower case letters x .

Example: Assume that a is randomly chosen from interval $[-1,1]$.

- $x = a^2$ is a continuous random variable.
- $x = \text{sgn}(a)$ is a discrete random variable.

Random Variables

In order to specify the probability measures used when dealing with random variables, it is often convenient to specify alternative functions from which the probability measure governing an experiment immediately follows.

- Probability Mass Function (PMF)
- Cumulative Distribution Function (CDF)
- Probability Density Function (PDF)

Random Variables: Probability Mass Function

A probability mass function (PMF) is a function $p_X : \Omega \rightarrow \mathbb{R}$ such that

$$p_X(x) \triangleq P(X = x).$$

In the case of discrete random variable, we use the notation $\text{Val}(X)$ for the set of possible values that the random variable X may assume. For example, if $X(\omega)$ is a random variable indicating the number of heads out of ten tosses of coin, then $\text{Val}(X) = \{0, 1, 2, \dots, 10\}$.

PMF Properties:

- $0 \leq p_X(x) \leq 1$.
- $\sum_{x \in \text{Val}(X)} p_X(x) = 1$.
- $\sum_{x \in A} p_X(x) = P(X \in A)$.

Random Variables: Probability Mass Function

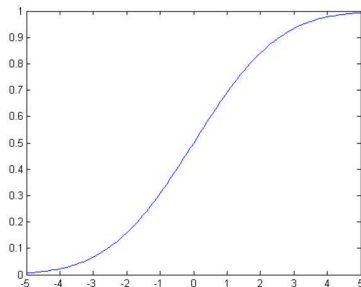
Example: Two independent tosses of a fair coin. Let X be the number of heads obtained in the two tosses. Determine the $\text{Val}(X)$ and the PMF of X .

Random Variables: Cumulative Distribution Function

A cumulative distribution function (CDF) is a function $F_X : \mathbb{R} \rightarrow [0, 1]$ which specifies a probability measure as,

$$F_X(x) \triangleq P(X \leq x).$$

By using this function one can calculate the probability of any event in $\mathcal{F}$



Random Variables: Cumulative Distribution Function

CDF Properties:

- $0 \leq F_X(x) \leq 1$.
- $\lim_{x \rightarrow -\infty} F_X(x) = 0$.
- $\lim_{x \rightarrow \infty} F_X(x) = 1$.
- $x \leq y \implies F_X(x) \leq F_X(y)$.
- $b \geq a, F_X(b) - F_X(a) = P(a < X \leq b)$.
- Right-Continuous: $\lim_{x \rightarrow x_0^+} F_X(x) = F_X(x_0)$.

Random Variables: Cumulative Distribution Function

Example: Determine the CDF of the discrete Random Variable X.

$$p_X(x) = \begin{cases} \frac{1}{4} & \text{if } x = 0 \\ \frac{3}{4} & \text{if } x = 2 \end{cases}$$

Random Variables: Probability Density Function

For some continuous random variables, the cumulative distribution function $F_X(x)$ is differentiable everywhere. In these cases, we define the Probability Density Function or PDF as the derivative of the CDF, i.e.,

$$f_X(x) \triangleq \frac{dF_X(x)}{dx}.$$

Note here, that the PDF for a continuous random variable may not always exist (i.e., if $F_X(x)$ is not differentiable everywhere).

$f_X(x)$ is a **valid PDF** if

- $f_X(x) \geq 0, \forall x$
- $\int_{-\infty}^{\infty} f_X(x) = 1$

Random Variables: Probability Density Function

PDF Properties:

- $f_X(x) \geq 0$.
- $\int_{-\infty}^{\infty} f_X(x) dx = 1$.
- $\int_{x \in A} f_X(x) dx = P(X \in A)$.

Random Variables: Probability Density Function

Example: Random Variable X has PDF $f_X(x)$.

Determine: (1) The constant c ? (2) CDF of X ? (3) $P(2 \leq X \leq 4)$?

$$f_X(x) = \begin{cases} \frac{c}{x^2} & \text{if } \frac{1}{2} \leq x \\ 0 & \text{otherwise} \end{cases}$$

Random Variables: Expectation of Discrete R.V.

Suppose that X is a discrete random variable with PMF $p_X(x)$, and $g : \mathbb{R} \rightarrow \mathbb{R}$ is an arbitrary function. In this case, $g(X)$ can be considered a random variable, and we define the expectation or expected value of $g(X)$ as

$$E[g(X)] \triangleq \sum_{x \in \text{Val}(X)} g(x)p_X(x).$$

Random Variables: Expectation of Continuous R.V.

If X is a continuous random variable with PDF $f_X(x)$, then the expected value of $g(X)$ is defined as,

$$E[g(X)] \triangleq \int_{-\infty}^{\infty} g(x)f_X(x)dx.$$

This is also known as LOTUS (Law Of The Unconscious Statistician).

Random Variables: Expectation

Expectation Properties:

- $E[a] = a$ for any constant $a \in \mathbb{R}$.
- $E[af(X)] = aE[f(X)]$ for any constant $a \in \mathbb{R}$.
- (Linearity of Expectation) $E[f(X) + g(X)] = E[f(X)] + E[g(X)]$.
- For a discrete random variable X , $E[1\{X = k\}] = P(X = k)$.

Random Variables: Expectation

Example: Suppose that $g(x) = 1\{x \in A\}$ for some subset $A \subseteq \Omega$. What is $E[g(X)]$?

Discrete case:

$$E[g(X)] = \sum_{x \in \text{Val}(X)} 1\{x \in A\} P_X(x) dx = \sum_{x \in A} P_X(x) dx = P(x \in A).$$

Continuous case:

$$E[g(X)] = \int_{-\infty}^{\infty} 1\{x \in A\} f_X(x) dx = \int_{x \in A} f_X(x) dx = P(x \in A).$$

Random Variables: Expectation

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Random Variables: Expectation

Example: Consider a continuous R.V. X with a uniform PDF over an interval $[a,b]$.

Determine: (1) CDF of X ? (2) Expectation of X ? (3) Expectation of X^2

Random Variables: Variance

The variance of a random variable X is a measure of how concentrated the distribution of a random variable X is around its mean. Formally, the variance of a random variable X is defined as

$$\text{Var}[X] \triangleq E[(X - E(X))^2] = E[X^2] - E[X]^2$$

Random Variables: Variance

Variance Properties:

- $\text{Var}[a] = 0$ for any constant $a \in \mathbb{R}$.
- $\text{Var}[af(X)] = a^2 \text{Var}[f(X)]$ for any constant $a \in \mathbb{R}$.

Random Variables: Variance

Example: Consider a continuous R.V. X with a uniform PDF over an interval $[a,b]$. Determine the variance of X .

Random Variables: Some Common Discrete R.V.s

- $X \sim \text{Bernoulli}(p)$ (where $0 \leq p \leq 1$): one if a coin with heads probability p comes up heads, zero otherwise.

$$p(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

Random Variables: Some Common Discrete R.V.s

- $X \sim \text{Binomial}(n, p)$ (where $0 \leq p \leq 1$): the number of heads in n independent flips of a coin with heads probability p .

$$p(x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Random Variables: Some Common Discrete R.V.s

- $X \sim \text{Geometric}(p)$ (where $p > 0$): the number of flips of a coin with heads probability p until the first heads.

$$p(x) = p(1 - p)^{x-1}$$

Random Variables: Some Common Discrete R.V.s

- $X \sim \text{Poisson}(\lambda)$ (where $\lambda > 0$): a probability distribution over the nonnegative integers used for modeling the frequency of rare events.

$$p(x) = e^{-\lambda} \frac{\lambda^x}{x!}$$

Random Variables: Some Common Continuous R.V.s

- $X \sim \text{Uniform}(a, b)$ (where $a < b$): equal probability density to every value between a and b on the real line.

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Random Variables: Some Common Continuous R.V.s

- $X \sim \text{Exponential}(\lambda)$ (where $\lambda > 0$): decaying probability density over the nonnegative reals.

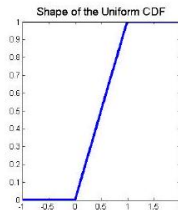
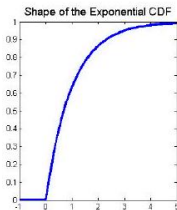
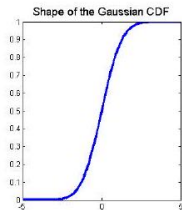
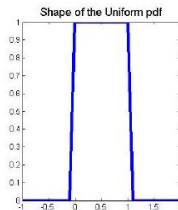
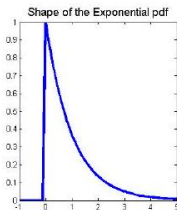
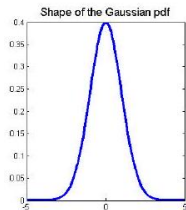
$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Random Variables: Some Common Continuous R.V.s

- $X \sim \text{Normal}(\mu, \sigma^2)$: also known as the Gaussian distribution

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

Random Variables: Some Common R.V.s



Random Variables: Some Common R.V.s

Distribution	PDF or PMF	Mean	Variance
Bernoulli (p)	$\begin{cases} p, \\ 1-p, \end{cases} \quad \text{if } x = 1$	p	$p(1-p)$
Binomial (n, p)	$\binom{n}{k} p^k (1-p)^{n-k}$ for $0 \leq k \leq n$	np	npq
Geometric (p)	$p(1-p)^{k-1}$ for $k = 1, 2, \dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson (λ)	$e^{-\lambda} \lambda^x / x!$ for $k = 1, 2, \dots$	λ	λ
Uniform (a, b)	$\frac{1}{b-a} \forall x \in (a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Gaussian (μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
Exponential (λ)	$\lambda e^{-\lambda x} \quad x \geq 0, \lambda > 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

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Two RVs: Joint and Marginal PMF

If X and Y are discrete random variables, then the joint probability mass function $p_{XY} : \mathbb{R} \times \mathbb{R} \rightarrow [0, 1]$ is defined by

$$p_{XY}(x, y) = P(X = x, Y = y).$$

Here, $0 \leq P_{XY}(x, y) \leq 1$ for all x, y , and $\sum_{x \in \text{Val}(X)} \sum_{y \in \text{Val}(Y)} P_{XY}(x, y) = 1$.

Two RVs: Joint and Marginal PMF

How does the joint PMF over two variables relate to the probability mass function for each variable separately? It turns out that

$$p_X(x) = \sum_y p_{XY}(x, y).$$

and similarly for $p_Y(y)$. In this case, we refer to $p_X(x)$ as the marginal probability mass function of X . In statistics, the process of forming the marginal distribution with respect to one variable by summing out the other variable is often known as "**marginalization**."

Two RVs: Joint and Marginal CDF

- Suppose that we have two random variables X and Y . One way to work with these two random variables is to consider each of them separately. If we do that we will only need $F_X(x)$ and $F_Y(y)$.
- But if we want to know about the values that X and Y assume simultaneously during outcomes of a random experiment, we require a more complicated structure known as the **joint** cumulative distribution function of X and Y , defined by $F_{XY}(x, y)$.

$$F_{XY}(x, y) = P(X \leq x, Y \leq y)$$

Two RVs: Joint and Marginal CDF

The joint CDF $F_{XY}(x, y)$ and the joint distribution functions $F_X(x)$ and $F_Y(y)$ of each variable separately are related by

$$F_X(x) = \lim_{y \rightarrow \infty} F_{XY}(x, y)$$

$$F_Y(y) = \lim_{x \rightarrow \infty} F_{XY}(x, y).$$

Here, we call $F_X(x)$ and $F_Y(y)$ the **marginal** cumulative distribution functions of $F_{XY}(x, y)$.

Two RVs: Joint and Marginal PDF

Let X and Y be two continuous random variables with joint distribution function F_{XY} . In the case that $F_{XY}(x, y)$ is everywhere differentiable in both x and y , then we can define the joint probability density function,

$$f_{XY}(x, y) = \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y}.$$

Two RVs: Joint and Marginal PDF

Let X and Y be two continuous random variables with joint distribution function F_{XY} . In the case that $F_{XY}(x, y)$ is everywhere differentiable in both x and y , then we can define the joint probability density function,

$$f_{XY}(x, y) = \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y}.$$

Like in the single-dimensional case, $f_{XY}(x, y) \neq P(X = x, Y = y)$, but rather

$$\iint_{(x,y) \in A} f_{XY}(x, y) dx dy = P((X, Y) \in A)$$

Two RVs: Joint and Marginal PDF

Similar to the discrete case, we define

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy,$$

as the marginal probability density function (or marginal density) of X , and similarly for $f_Y(y)$.

Two RVs: Conditional PMF

Conditional distributions seek to answer the question, what is the probability distribution over Y , when we know that X must take on a certain value x ?

In the **discrete** case, the **conditional PMF** of Y given X is:

$$p_{Y|X}(y | x) = P(Y = y | X = x) = \frac{P(Y = y \cap X = x)}{P(X = x)} = \frac{p_{XY}(x, y)}{p_X(x)},$$

Alternatively,

$$p_{XY}(x, y) = p_X(x)p_{Y|X}(y | x)$$

Two RVs: Conditional PDF

In the **continuous** case, the **conditional PDF** of Y given X is:

$$f_{Y|X}(y | x) = \frac{f_{XY}(x, y)}{f_X(x)},$$

Alternatively,

$$f_{XY}(x, y) = f_X(x)f_{Y|X}(y | x)$$

Two RVs: Bayes's Rule

A useful formula that often arises when trying to derive expression for the conditional probability of one variable given another, is Bayes's rule.

In the case of **discrete** random variables X and Y ,

$$P_{Y|X}(y | x) = \frac{P_{XY}(x, y)}{P_X(x)} = \frac{P_{X|Y}(x | y) P_Y(y)}{\sum_{y' \in \text{Val}(Y)} P_{X|Y}(x | y') P_Y(y')}.$$

In the case of **continuous** random variables X and Y ,

$$f_{Y|X}(y | x) = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{f_{X|Y}(x | y) f_Y(y)}{\int_{-\infty}^{\infty} f_{X|Y}(x | y') f_Y(y') dy'}.$$

Two RVs: Example

Example: Assume that the probability that it rains in Ann Arbor given that the temperature is above $80^{\circ}F$ is 0.6. When the temperature is below $80^{\circ}F$, the probability of raining is only 0.1. The probability that the temperature is above $80^{\circ}F$ is 0.25.

- Given that we observed rain in Ann Arbor today, what is the probability that the temperature was high too?

Two RVs: Example

Example: Two continuous Random Variables X and Y have joint PDF as below (c is an unknown constant). Please calculate $P(Y \geq X)$:

$$f_{XY}(x, y) = \begin{cases} c & \text{if } 0 \leq x \leq 5, 0 \leq y \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

Two RVs: Independence

Two random variables X and Y are **independent** if $F_{XY}(x, y) = F_X(x)F_Y(y)$ for all values of x and y . Equivalently,

- For discrete random variables, $p_{XY}(x, y) = p_X(x)p_Y(y)$ for all $x \in \text{Val}(X)$, $y \in \text{Val}(Y)$
- For discrete random variables, $p_{Y|X}(y | x) = p_Y(y)$ whenever $p_X(x) \neq 0$ for all $y \in \text{Val}(Y)$
- For continuous random variables, $f_{XY}(x, y) = f_X(x)f_Y(y)$ for all $x, y \in \mathbb{R}$.
- For continuous random variables, $f_{Y|X}(y | x) = f_Y(y)$ whenever $f_X(x) \neq 0$ for all $y \in \mathbb{R}$.

Two RVs: Expectation and Covariance

Suppose that we have two discrete random variables X, Y and $g : \mathbf{R}^2 \longrightarrow \mathbf{R}$ is a function of these two random variables. Then the **expectation** of g is defined in the following way,

$$E[g(X, Y)] \triangleq \sum_{x \in \text{Val}(X)} \sum_{y \in \text{Val}(Y)} g(x, y) p_{XY}(x, y).$$

For continuous random variables X, Y , the analogous expression is

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{XY}(x, y) dx dy.$$

Two RVs: Expectation and Covariance

We can use the concept of expectation to study the relationship of two random variables with each other. In particular, the **covariance** of two random variables X and Y is defined as

$$\text{Cov}[X, Y] \triangleq E[(X - E[X])(Y - E[Y])]$$

Using an argument similar to that for variance, we can rewrite this as,

$$\begin{aligned}\text{Cov}[X, Y] &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY - XE[Y] - YE[X] + E[X]E[Y]] \\ &= E[XY] - E[X]E[Y] - E[Y]E[X] + E[X]E[Y] \\ &= E[XY] - E[X]E[Y].\end{aligned}$$

Two RVs: Expectation and Covariance

We can use the concept of expectation to study the relationship of two random variables with each other. In particular, the **covariance** of two random variables X and Y is defined as

$$\text{Cov}[X, Y] \triangleq E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y]$$

Here, the key step in showing the equality of the two forms of covariance is in the third equality, where we use the fact that $E[X]$ and $E[Y]$ are actually constants which can be pulled out of the expectation. When $\text{Cov}[X, Y] = 0$, we say that X and Y are **uncorrelated**.

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Multiple RV: Definitions

We can define the joint CDF of $X_1, X_2, \dots, X_n$, the joint PDF of $X_1, X_2, \dots, X_n$, the marginal PDF of X_1 , and the conditional PDF of X_1 given $X_2, \dots, X_n$, as

$$F_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n)$$

$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = \frac{\partial^n F_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n)}{\partial x_1 \dots \partial x_n}$$

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) dx_2 \dots dx_n$$

$$f_{X_1|X_2, \dots, X_n}(x_1 | x_2, \dots, x_n) = \frac{f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n)}{f_{X_2, \dots, X_n}(x_2, \dots, x_n)}$$

To calculate the probability of an event $A \subseteq \mathbb{R}^n$ we have,

$$P((x_1, x_2, \dots, x_n) \in A) = \int_{(x_1, x_2, \dots, x_n) \in A} f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) dx_1 dx_2 \dots dx_n$$

Multiple RV: Multivariate gaussian Distribution

One particularly important example of a probability distribution over random vectors X is called the **multivariate Gaussian** or **multivariate normal** distribution. A random vector $X \in \mathbb{R}^n$ is said to have a multivariate normal (or Gaussian) distribution with mean $\mu \in \mathbb{R}^n$ and covariance matrix $\Sigma \in \mathbb{S}_{++}^n$ (where $\mathbb{S}_{++}^n$ refers to the space of symmetric positive definite $n \times n$ matrices)

$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right).$$

We write this as $X \sim \mathcal{N}(\mu, \Sigma)$. Notice that in the case $n = 1$, this reduces the regular definition of a normal distribution with mean parameter μ_1 and variance Σ_{11} .

Questions

Questions